1 Introduction

This experiment compares full fine-tuning and LoRA (Low-Rank Adaptation), a parameter-efficient fine-tuning (PEFT) method, using the **Qwen2-0.5B** language model (∼490 M parameters) on the **IMDB** sentiment analysis dataset (binary positive/negative classification, 50 k samples).

2 Methodology

1. **Dataset & Preprocessing:** IMDB dataset via Hugging Face Datasets was used in this experiment. Text is tokenized with the Qwen2 tokenizer. A random subset of 10,000 training samples and 2,000 test samples (seed = 42) is used to keep training feasible on a single GPU on Google Colab.

2. **Full Fine-Tuning:** All ∼490 M parameters are updated, and loaded the model in `bfloat16` precision and enable gradient checkpointing to reduce memory usage.

3. **LoRA (PEFT):** Only low-rank adapter matrices are trained while the base model weights are frozen. The LoRA configuration targets the attention projection layers (`q_proj`, `k_proj`, `v_proj`, `o_proj`) with rank $r = 8$, scaling factor $\alpha = 16$, and dropout 0.1.

4. **Hyperparameters:** Both methods use the same training setup for a fair comparison: learning rate $5{\times}10^{-5}$, per-device batch size 8 with gradient accumulation steps 2 (effective batch size 16), 1 training epoch, `bfloat16` mixed precision, and the AdamW optimizer.

3 Results

Table 1 summarizes the performance and efficiency of each method.

Method	**Accuracy (%)**	**Training Time (min)**	**Peak GPU Memory (GB)**
Full Fine-Tuning	94.4	63.8	6.0
LoRA (PEFT)	91.8	53.2	3.1

Table 1: Comparison of full fine-tuning and LoRA on IMDB sentiment analysis with Qwen2-0.5B.

Full fine-tuning updates all ∼490 M parameters and achieves a slightly higher accuracy, but at the cost of greater GPU memory consumption and longer training time. LoRA, by contrast, trains only a small fraction of parameters (the low-rank adapters appended to each attention layer) and therefore requires significantly less memory and time while achieving competitive accuracy.

4 Conclusion

Both methods achieve strong performance on IMDB binary sentiment classification, demonstrating that even a 0.5 B-parameter model can effectively learn this task in a single epoch. Full fine-tuning yields the highest accuracy by updating every weight in the network, but LoRA delivers comparable results at a fraction of the computational cost. For resource-constrained settings, such as free-tier cloud GPUs, LoRA provides an excellent trade-off between accuracy and efficiency, making it the more practical choice for fine-tuning LLMs.